

A Mixed-Base Counterexample to Stanley's Rationality Question

Exact coefficient counts, irrational thresholds,
and a natural boundary

Research report prepared for Vladimir Reshetnikov

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Abstract

Richard Stanley asked whether a positive integer sequence tending to infinity and having a rational generating function necessarily produces rational generating functions for the numbers of coefficients equal to a fixed integer in its associated finite products. We give a counterexample using the interleaved sequences 2^j and 3^j . For $P_N(x) = \prod_{i=1}^N (1 + x^{G_i})$, with $G_{2j+1} = 2^j$ and $G_{2j+2} = 3^j$, the number a_n of coefficients equal to 1 in P_{2n} is

$$a_n = 2^{n - \lfloor (n+1) \log_3 2 \rfloor} + \mathbf{1}_{\{n=1\}}.$$

The proof turns polynomial coefficients into interval-covering multiplicities and then counts gaps created by ternary carries. A finite-field argument proves nonrationality without any transcendence assumption about the exponential growth constant. A separate analytic argument proves a natural boundary, hence non-D-finiteness and the absence of polynomial-coefficient linear recurrences. Finally, for a two-base family with digits $0, \dots, a-1$, we prove that rationality holds exactly when the bases are multiplicatively dependent. Exact-integer verification code, sequence data, and reproducible figures accompany the proofs.

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1 The problem and the result

In a MathOverflow question dated September 23, 2022, Richard Stanley considers a sequence of positive integers $(G_N)_{N \geq 1}$ satisfying

$$G_N \longrightarrow \infty, \quad \sum_{N \geq 1} G_N t^N \in \mathbb{Q}(t). \quad (1.1)$$

For a fixed integer $r \geq 1$, put

$$P_N(x) = \prod_{i=1}^N (1 + x^{G_i} + \cdots + x^{rG_i}), \quad P_0(x) = 1. \quad (1.2)$$

For fixed $k \geq 1$, let $v_k(N)$ be the number of exponents $s \geq 0$ for which $[x^s]P_N(x) = k$. The question asks whether $\sum_{N \geq 0} v_k(N)t^N$ is rational and, if not universally, which sequences G_N have this property [1].

The present report gives a negative answer to the universal assertion, already for $r = k = 1$. The source question requires divergence to infinity, *not monotonicity*. That distinction matters: the counterexample below is not monotone.

1.1 An explicit counterexample

For every $j \geq 0$, define

$$G_{2j+1} = 2^j, \quad G_{2j+2} = 3^j. \quad (1.3)$$

Thus the beginning of the exponent sequence is

$$1, 1, 2, 3, 4, 9, 8, 27, 16, 81, 32, 243, \dots$$

Its generating function is visibly rational:

$$\sum_{N \geq 1} G_N t^N = \frac{t}{1-2t^2} + \frac{t^2}{1-3t^2} = \frac{t + t^2 - 3t^3 - 2t^4}{1-5t^2+6t^4}. \quad (1.4)$$

Equivalently,

$$G_{N+4} = 5G_{N+2} - 6G_N \quad (N \geq 1). \quad (1.5)$$

Both parity subsequences tend to infinity, so (1.1) holds. Write

$$v(N) = v_1(N), \quad a_n = v(2n), \quad b_n = v(2n+1), \quad A(z) = \sum_{n \geq 0} a_n z^n, \quad F(t) = \sum_{N \geq 0} v(N)t^N. \quad (1.6)$$

The variable x always records polynomial exponents; z and t record product lengths. In particular, A is a section of F , not a specialization of the product variable x .

Theorem 1.1 (Counterexample and analytic strengthening). *Set $\alpha = \log_3 2$ and $R = 2^{\alpha-1}$. For the exponents (1.3),*

$$a_n = 2^{n - \lfloor (n+1)\alpha \rfloor} + \mathbf{1}_{\{n=1\}} \quad (n \geq 0). \quad (1.7)$$

Neither $A(z)$ nor $F(t)$ is rational. In fact, $|z| = R$ is a natural boundary for A , and $|t| = \sqrt{R}$ is a natural boundary for F . Consequently both functions are non-D-finite and transcendental over their rational function fields, and neither coefficient sequence is P-recursive.

The elementary counterexample is complete by Section 4. Sections 5–7 establish the exact floor formula and the stronger analytic conclusions. Section 8 proves a necessary-and-sufficient condition for rationality in a broader two-base family.

1.2 Source status and scope of the claim

The source page displayed no posted answer when inspected on September 19, 2026. Targeted searches for the question title, its identifier, and mixed-base counterexamples did not reveal an existing resolution. This is a limited literature audit, not a proof of priority. The mathematical assertions here are supported by the proofs below; the report has not been independently refereed or formally verified in a proof assistant.

Stanley's related work on rational generating functions for moments of polynomial coefficients supplies useful context [2]. Those moment questions and the fixed-multiplicity question (1.2) are different. This report does not claim to contradict a theorem or settle the other conjectures in that paper.

2 From polynomial coefficients to interval coverage

The basic mechanism is that one half of the product creates a solid interval of monomials, while the other half creates a sparse set of translates. Counting coefficients equal to one then becomes a geometric counting problem on the integer line.

2.1 A general unique-coverage identity

Let $s_0 < s_1 < \dots < s_{M-1}$ be integer starting points, $M \geq 2$, and let $L \geq 1$ be an integer. Consider the half-open intervals

$$I_i = [s_i, s_i + L), \quad 0 \leq i < M.$$

Only their integer points are counted. Write $d_i = s_{i+1} - s_i$ for $0 \leq i < M - 1$.

Lemma 2.1 (Unique coverage). *The number of integers belonging to exactly one of the intervals I_i is*

$$\min(d_0, L) + \min(d_{M-2}, L) + \sum_{i=1}^{M-2} \max(0, \min(d_{i-1}, L) + \min(d_i, L) - L). \quad (2.1)$$

The sum is empty when $M = 2$.

Proof. An integer in I_0 belongs to no other interval exactly when it is smaller than s_1 . There are $\min(d_0, L)$ such integers. The last interval similarly contributes $\min(d_{M-2}, L)$.

Now fix an internal interval I_i . Because all intervals have the same length, the immediate predecessor has the rightmost endpoint among all previous intervals, and the immediate successor has the leftmost starting point among all later ones. An integer in I_i therefore belongs to no other interval exactly when it lies in

$$[\max(s_i, s_{i-1} + L), \min(s_i + L, s_{i+1})).$$

The endpoints are integers. Its number of integer points is its length when positive, and zero otherwise. Subtracting s_i from both endpoints gives

$$\max(0, \min(L, d_i) - \max(0, L - d_{i-1})),$$

which is the corresponding summand in (2.1). Each uniquely covered point belongs to exactly one interval, so these contributions are disjoint and exhaustive. $\square$

Corollary 2.2 (Isolated large gaps). *Suppose $L \geq 2$, the first and last gaps are 1, and no two gaps greater than 1 are adjacent. If H gaps have size at least L , the number of uniquely covered integers is*

$$2 + 2H. \quad (2.2)$$

Proof. The two boundary terms in (2.1) are 1. For an internal interval at least one neighboring gap is 1. If the other gap is g , the contribution is

$$\max(0, 1 + \min(g, L) - L).$$

As g and L are integers, this is 1 exactly when $g \geq L$, and is zero otherwise. Each gap of size at least L is nonunit, is not a boundary gap, and borders two internal intervals. It therefore contributes twice. $\square$

2.2 The rectangular binary-ternary product

For $m, n \geq 0$, define

$$Q_{m,n}(x) = \prod_{j=0}^{m-1} (1 + x^{2^j}) \prod_{j=0}^{n-1} (1 + x^{3^j}). \quad (2.3)$$

The first product enumerates all binary expansions of length m , uniquely, so

$$\prod_{j=0}^{m-1} (1 + x^{2^j}) = 1 + x + \dots + x^{2^m-1}. \quad (2.4)$$

The second product has coefficient 1 precisely at the points of

$$S_n = \left\{ \sum_{j=0}^{n-1} \epsilon_j 3^j : \epsilon_j \in \{0, 1\} \right\}, \quad |S_n| = 2^n. \quad (2.5)$$

Ternary expansions are unique, so there is no multiplicity within S_n . Consequently

$$[x^s]Q_{m,n}(x) = \#\{u \in S_n : u \leq s < u + 2^m\}. \quad (2.6)$$

This identity reduces the coefficient problem exactly, not approximately, to Lemma 2.1 with interval length $L = 2^m$.

3 Ternary carries and exact counts

3.1 The complete gap distribution

The elements of S_n are sorted by interpreting the binary digit strings $\epsilon_{n-1} \dots \epsilon_0$ as ternary strings. This preserves their order: changing the highest differing digit outweighs all lower digits.

Lemma 3.1 (Ternary support gaps). *For $n \geq 1$, the gaps between consecutive elements of S_n have sizes*

$$g_t = \frac{3^t + 1}{2}, \quad 0 \leq t < n. \quad (3.1)$$

The gap g_t occurs 2^{n-1-t} times. The first and last gaps are 1, and every pair of adjacent gaps contains a gap of size 1.

Proof. To move to the next binary digit string, turn a string of t trailing ones into zeroes and turn the next zero into a one. The corresponding change in the ternary value is

$$3^t - \sum_{j=0}^{t-1} 3^j = 3^t - \frac{3^t - 1}{2} = \frac{3^t + 1}{2}.$$

For a carry of length t , the t trailing digits and the next digit are fixed, while the $n - 1 - t$ higher digits are arbitrary. This gives the stated multiplicity. A nontrivial carry resets the last digit to zero, so the next step changes only that digit and has gap 1. The step immediately before a nontrivial carry is also a unit step. The first and last steps both change only the last digit. These observations establish the adjacency and boundary claims. $\square$

The multiplicities sum to $2^n - 1$, as required. Although the largest gap grows exponentially, gaps larger than one remain separated by unit gaps. That separation is what makes the number of coefficients equal to one simple.

3.2 An exact integer threshold

Define

$$T(m) = \min\{t \geq 0 : 3^t \geq 2^{m+1} - 1\}. \quad (3.2)$$

For $m \geq 1$, this is at least 1. Let $U(m, n)$ denote the number of coefficients equal to one in $Q_{m,n}$.

Theorem 3.2 (Rectangular count). *For $m, n \geq 1$,*

$$U(m, n) = 2^{1+\max(0, n-T(m))}. \quad (3.3)$$

The boundary cases are

$$U(m, 0) = 2^m, \quad U(0, n) = 2^n. \quad (3.4)$$

In particular $U(0, 0) = 1$.

Proof. A gap g_t is at least 2^m exactly when $t \geq T(m)$. If $T(m) < n$, Lemma 3.1 gives

$$H = \sum_{t=T(m)}^{n-1} 2^{n-1-t} = 2^{n-T(m)} - 1.$$

If $T(m) \geq n$, then $H = 0$. Apply Corollary 2.2 and simplify $2 + 2H$. When one of the products is empty, the other has all its nonzero coefficients equal to one, proving (3.4). $\square$

Since $P_{2n} = Q_{n,n}$ and $P_{2n+1} = Q_{n+1,n}$, this already determines the entire sequence $v(N)$ by integer arithmetic. For $n \geq 1$, the inequality

$$3^n + 1 \geq 2^{n+1} \quad (3.5)$$

holds by induction, so $T(n) \leq n$. Thus

$$a_0 = 1, \quad a_n = 2^{n-T(n)+1} \quad (n \geq 1). \quad (3.6)$$

The first values are given in Table 1. In the table $T(0) = 0$; formula (3.6) has its stated separate case at $n = 0$.

Table 1: Exact counts on the even subsequence.

n	$T(n)$	a_n	n	$T(n)$	a_n	n	$T(n)$	a_n
0	0	1	7	6	4	14	10	32
1	1	2	8	6	8	15	11	32
2	2	2	9	7	8	16	11	64
3	3	2	10	7	16	17	12	64
4	4	2	11	8	16	18	12	128
5	4	4	12	9	16	19	13	128
6	5	4	13	9	32	20	14	128

3.3 A fully visible example

Take $m = n = 5$. The interval length is 32, while the gap sizes are 1, 2, 5, 14, 41. Only the gap 41 reaches the threshold; it occurs once. Therefore $U(5, 5) = 2 + 2 = 4$.

Concretely, the support S_5 splits into a block ending at $1 + 3 + 9 + 27 = 40$ and a block beginning at 81. The first block's final interval ends just after 71, while the second block starts at 81. The four uniquely covered integers are

$$0, \quad 71, \quad 81, \quad 152. \quad (3.7)$$

The outer two come from the global boundaries, and the middle two flank the unique large gap. Direct multiplication of P_{10} gives precisely these coefficient-one positions; Figure 1 displays all coefficients, including the zero coefficients between the two blocks.

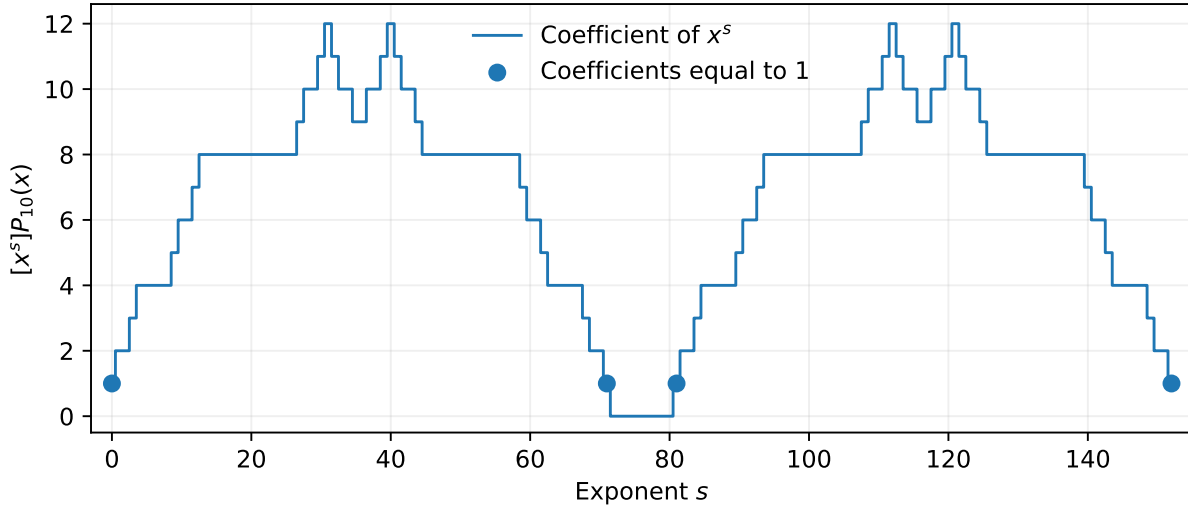


Figure 1: The coefficients of $P_{10} = Q_{5,5}$. The four marked values occur at the positions in (3.7). The maximum coefficient is 12; the sum of all coefficients is $2^{10} = 1024$.

4 An elementary obstruction to rationality

The exact count is a power of two. Its exponent grows at an irrational average speed but changes only by zero or one at each step. Rational generating functions cannot exhibit this particular combination.

4.1 Rational recurrences modulo a prime

A sequence is called *C-finite* when, from some index onward, it satisfies a linear recurrence with constant coefficients. An ordinary generating function over $\mathbb{Q}$ is rational if and only if its coefficients are C-finite over $\mathbb{Q}$: multiplying the series by its denominator gives the recurrence, and summing an eventual recurrence gives a polynomial numerator divided by a polynomial denominator.

There is no change in this conclusion if rationality is initially allowed over $\mathbb{C}$ for a series with rational coefficients. For a fixed denominator degree and a fixed numerator bound, the coefficient identities are linear equations with rational entries in finitely many unknown denominator coefficients, with the constant denominator coefficient normalized to one. A finite subsystem spans all the augmented rows. If a complex solution exists, rational Gaussian elimination gives a rational solution to that subsystem and hence to all the identities. Thus the rational function can be taken over $\mathbb{Q}$.

Lemma 4.1 (Finite ratios and irrational exponent slope). *Let $b \geq 2$ and $C \geq 1$ be integers. Suppose an integer sequence (u_n) has, for all sufficiently large n , the form*

$$u_n = Cb^{e_n}, \quad e_n \in \mathbb{Z}_{\geq 0}, \quad e_{n+1} - e_n \in \{0, 1\}.$$

If $\lim_{n \rightarrow \infty} e_n/n$ exists and is irrational, then $\sum_{n \geq 0} u_n z^n$ is not rational.

Proof. Suppose the generating function were rational. Then for some $d \geq 1$ and rational constants $c_0, \dots, c_{d-1}$,

$$u_{n+d} = c_{d-1}u_{n+d-1} + \dots + c_0u_n \tag{4.1}$$

for every sufficiently large n . Choose a prime p dividing neither $Cb(b-1)$ nor any denominator of the c_j . Only finitely many primes are excluded.

Reduce (4.1) modulo p . The state vector $(u_n, \dots, u_{n+d-1}) \bmod p$ belongs to the finite set $\mathbb{F}_p^d$ and has a deterministic successor. Its forward orbit, and hence $u_n \bmod p$, is eventually periodic. No invertibility of the transition is needed.

For large n , all $u_n \bmod p$ are nonzero. The quotient $u_{n+1}/u_n \bmod p$ is therefore eventually periodic as well. Its actual integer value is either 1 or b , and these have distinct residues modulo p . The periodic residue quotient thus uniquely determines the actual quotient. Consequently $e_{n+1} - e_n$ is eventually periodic.

If its eventual period has length L and contains J ones, summing the increments yields $e_n = (J/L)n + O(1)$. The limit e_n/n must be the rational number J/L , contrary to the hypothesis. $\square$

4.2 Applying the obstruction

By its definition, $T(n)$ is nondecreasing. For $n \geq 1$, if $3^{T(n)} \geq 2^{n+1} - 1$, then

$$3^{T(n)+1} \geq 3(2^{n+1} - 1) \geq 2^{n+2} - 1.$$

Therefore

$$T(n+1) - T(n) \in \{0, 1\} \quad (n \geq 1). \tag{4.2}$$

Moreover, the two adjacent powers bracketing the threshold give

$$T(n) = n \log_3 2 + O(1). \tag{4.3}$$

For $n \geq 1$ put $e_n = n - T(n) + 1$. Equations (3.6), (4.2), and (4.3) show that

$$a_n = 2^{e_n}, \quad e_{n+1} - e_n \in \{0, 1\}, \quad \frac{e_n}{n} \longrightarrow 1 - \log_3 2.$$

The last limit is irrational. Indeed, a rational equality $\log_3 2 = p/q$ with positive integers p, q would imply $2^q = 3^p$, contradicting unique prime factorization. Lemma 4.1 proves that A is not rational.

Finally, the even section of a rational series is rational. One way to see this is to represent an eventual recurrence by a finite companion matrix: sampling at even indices replaces the matrix by its square and again gives a constant-coefficient recurrence, by the characteristic polynomial. Equivalently,

$$\frac{F(t) + F(-t)}{2} = A(t^2),$$

and an even rational function of t is a rational function of t^2 . Thus rationality of F would imply rationality of A , a contradiction. This proves the negative answer to the universal rationality question.

Remark 4.2 (What the argument does not assume). The irrationality of an exponential growth constant by itself would not rule out rationality: coefficients of a rational function can have irrational algebraic exponential growth. The proof instead uses an irrational *frequency of discrete exponent increments* and finite-state arithmetic modulo a prime. It makes no algebraicity or transcendence claim about $2^{1-\log_3 2}$.

5 The floor formula and the odd indices

5.1 Removing the near-integer ambiguity

The threshold (3.2) involves $2^{n+1} - 1$, not 2^{n+1} . The difference is important for the initial terms and must not be silently dropped.

Proposition 5.1 (Exact floor expression). *For $n \geq 2$,*

$$T(n) = \lceil (n+1)\alpha \rceil = \lfloor (n+1)\alpha \rfloor + 1, \quad \alpha = \log_3 2. \quad (5.1)$$

Consequently (1.7) holds for every $n \geq 0$.

Proof. Let $M = \lceil (n+1)\alpha \rceil$ be the least integer such that $3^M \geq 2^{n+1}$. The number $(n+1)\alpha$ is irrational, so the inequality is strict and the ceiling is one more than the floor. Since the smaller threshold in (3.2) is $2^{n+1} - 1$, we have $T(n) \leq M$. If $T(n) < M$, an integer power of three is at least $2^{n+1} - 1$ but smaller than 2^{n+1} . It must equal $2^{n+1} - 1$. For $n \geq 2$, the latter is 7 modulo 8, whereas a power of three is 1 or 3 modulo 8. This is impossible.

Substitute (5.1) into (3.6). At $n = 0$, $2^{0-\lfloor \alpha \rfloor} = 1 = a_0$. At $n = 1$, $2^{1-\lfloor 2\alpha \rfloor} = 1$ while $a_1 = 2$, producing exactly the correction $\mathbf{1}_{\{n=1\}}$. $\square$

For $n \geq 2$ the logarithmic exponent increments can also be written as

$$\log_2 \frac{a_{n+1}}{a_n} = 1 - (\lfloor (n+2)\alpha \rfloor - \lfloor (n+1)\alpha \rfloor). \quad (5.2)$$

This is a binary sequence driven by an irrational rotation. Its long-run frequency of ones is $1 - \alpha$, either by telescoping the floors or by the equidistribution argument in the next section. In particular, the proportion of indices at which a_n doubles tends to $1 - \alpha$.

5.2 The entire product-length sequence

The rectangular formula also determines the odd indices without a new counting argument.

Proposition 5.2 (Even–odd identity). *For every $n \geq 0$,*

$$b_n = \frac{a_{n+1}}{2} + \mathbf{1}_{\{0 \leq n \leq 3\}}. \quad (5.3)$$

Thus

$$F(t) = \frac{(2t+1)A(t^2) - 1}{2t} + t + t^3 + t^5 + t^7, \quad (5.4)$$

where the apparent singularity at $t = 0$ is removable.

Proof. For $n \geq 1$, Theorem 3.2 gives $b_n = 2^{1+\max(0, n-T(n+1))}$, while $a_{n+1}/2 = 2^{n+1-T(n+1)}$. The bound $T(n+1) \leq n+1$ shows that these agree unless $T(n+1) = n+1$, in which case their values are 2 and 1. Directly, $T(2) = 2$, $T(3) = 3$, and $T(4) = 4$. For $n \geq 4$ the inequality $3^n \geq 2^{n+2} - 1$ follows from its case $n = 4$ by induction; hence $T(n+1) \leq n$. At $n = 0$ the required identity is $b_0 = 2 = a_1/2 + 1$.

Writing $B(z) = \sum_{n \geq 0} b_n z^n$, summation gives

$$B(z) = \frac{A(z) - 1}{2z} + 1 + z + z^2 + z^3.$$

Substitute this into $F(t) = A(t^2) + tB(t^2)$ to obtain (5.4). $\square$

The initial values, indexed by the actual number of factors, are

$$(v(N))_{N=0}^{22} = (1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 4, 2, 4, 2, 4, 2, 4, 8, 4, 8, 8, 16, 8, 16). \quad (5.5)$$

The drops at odd indices are real; adding a factor can turn many coefficients that were one into larger coefficients. The even subsequence is monotone, but the full coefficient-one sequence need not be.

6 Asymptotics with persistent oscillation

Set

$$\lambda = 2^{1-\alpha}, \quad R = \lambda^{-1}, \quad h(u) = 2^{u-\alpha} \quad (0 \leq u < 1), \quad u_n = \{(n+1)\alpha\}. \quad (6.1)$$

Numerically,

$$\alpha \approx 0.630929753571, \quad \lambda \approx 1.291520234330, \quad R \approx 0.774281326315.$$

These decimal approximations play no role in the exact computations or proofs. Proposition 5.1 gives the exact representation

$$a_n = \lambda^n h(u_n) + \mathbf{1}_{\{n=1\}}. \quad (6.2)$$

Thus the relative fluctuations do not shrink with n ; they are the values of a fixed function sampled along an irrational rotation.

6.1 The equidistribution input

We use the following elementary consequence of the geometric-sum formula.

Lemma 6.1 (Irrational rotation averages). *If β is irrational and f is Riemann integrable on $[0, 1]$, with its values interpreted periodically, then for every real γ ,*

$$\lim_{M \rightarrow \infty} \frac{1}{M} \sum_{n=0}^{M-1} f(\{n\beta + \gamma\}) = \int_0^1 f(u) du. \quad (6.3)$$

Proof. For each nonzero integer j ,

$$\frac{1}{M} \sum_{n=0}^{M-1} e^{2\pi i j (n\beta + \gamma)} = \frac{e^{2\pi i j \gamma}}{M} \frac{1 - e^{2\pi i j M \beta}}{1 - e^{2\pi i j \beta}} \rightarrow 0.$$

For $j = 0$ the average is 1. Therefore (6.3) holds for trigonometric polynomials. Uniform approximation by trigonometric polynomials extends it to continuous periodic functions. Continuous upper and lower approximations to an interval indicator then give its average as the interval length: the approximations can differ only in arbitrarily short neighborhoods of the endpoints. Finite step functions follow. Finally, upper and lower Riemann sums approximate every real Riemann integrable f with arbitrarily small integral gap. Apply the same argument to the real and imaginary parts for complex-valued f . $\square$

Proposition 6.2 (Growth and fluctuation law). *The sequence a_n satisfies*

$$\lim_{n \rightarrow \infty} a_n^{1/n} = \lambda, \quad a_n = \Theta(\lambda^n). \quad (6.4)$$

The set of limit points of a_n / λ^n is the whole interval

$$[2^{-\alpha}, 2^{1-\alpha}]. \quad (6.5)$$

In particular there is no constant C for which $a_n \sim C\lambda^n$. Its normalized Cesàro mean does exist:

$$\lim_{M \rightarrow \infty} \frac{1}{M} \sum_{n=0}^{M-1} \frac{a_n}{\lambda^n} = \frac{2^{-\alpha}}{\log 2}. \quad (6.6)$$

The full sequence has root growth $\lim_{N \rightarrow \infty} v(N)^{1/N} = \sqrt{\lambda}$.

Proof. Outside the single corrected index, (6.2) lies between the two fixed positive bounds $2^{-\alpha} \lambda^n$ and $2^{1-\alpha} \lambda^n$. This proves (6.4). Lemma 6.1 implies that the fractional parts u_n visit every subinterval of $[0, 1)$ infinitely often. The strictly increasing function h therefore gives every interior point of (6.5) as a subsequential limit; approaching 0 and 1 gives the endpoints. No other limits are possible. Equation (6.6) follows by integrating h , and the finite exceptional term vanishes in the average. Finally, (5.3) gives the same root growth on both parities of $v(N)$. $\square$

7 Dense singularities and a natural boundary

The fluctuation law does more than prevent a constant leading asymptotic. Every Fourier mode of the profile h is nonzero. Each mode produces a boundary singularity, and the corresponding points are dense on the circle of convergence.

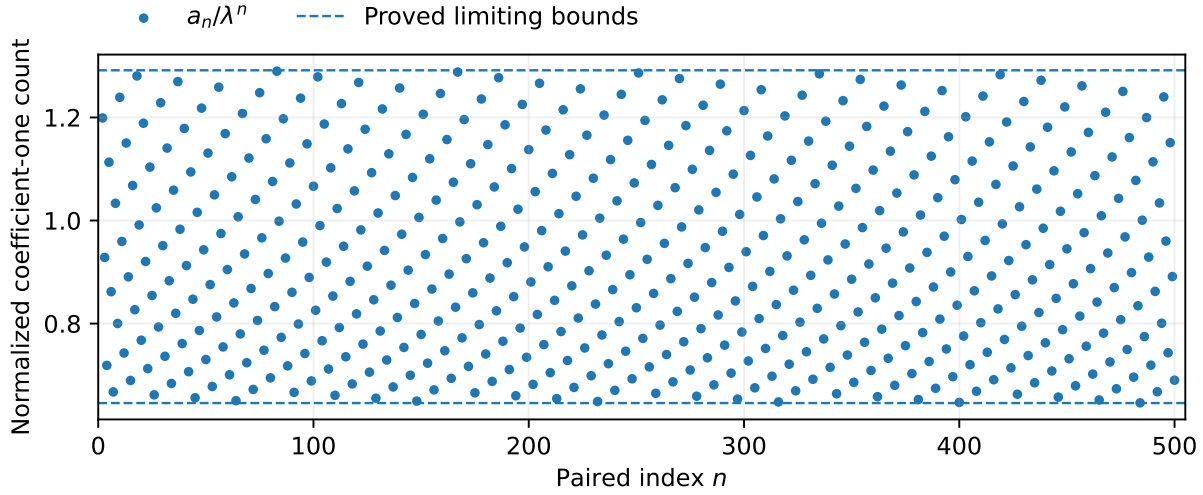


Figure 2: The normalized counts a_n/λ^n for $2 \leq n \leq 500$. The dashed lines are the limiting bounds in (6.5). The nonvanishing oscillation is proved by irrational rotation, not inferred from this plot. Floating-point arithmetic is used only to draw the figure.

7.1 A precise Abelian lemma

Lemma 7.1 (Cesàro means imply radial limits). *Let (c_n) be a bounded complex sequence. If $M^{-1} \sum_{n=0}^{M-1} c_n \rightarrow c$, then*

$$\lim_{\rho \uparrow 1} (1 - \rho) \sum_{n \geq 0} c_n \rho^n = c. \quad (7.1)$$

Proof. Put $S_N = \sum_{n=0}^N c_n$. Absolute convergence for $0 < \rho < 1$ gives

$$\sum_{n \geq 0} c_n \rho^n = (1 - \rho) \sum_{N \geq 0} S_N \rho^N.$$

The hypothesis is $S_N = (N + 1)c + o(N + 1)$. Since $\sum_{N \geq 0} (N + 1)\rho^N = (1 - \rho)^{-2}$, the principal term contributes c after multiplying by $1 - \rho$. For the error, fix $\varepsilon > 0$ and choose N_0 so that its absolute value is at most $\varepsilon(N + 1)$ for $N \geq N_0$. The tail contributes at most ε , and the finitely many initial terms tend to zero. This proves (7.1). $\square$

7.2 Nonzero radial amplitudes at a dense set

Theorem 7.2 (Explicit dense boundary singularities). *For each $k \in \mathbb{Z}$, put $\omega_k = e^{-2\pi i k \alpha}$. Then*

$$\lim_{\rho \uparrow 1} (1 - \rho) A(R\rho\omega_k) = \frac{2^{-\alpha} e^{2\pi i k \alpha}}{\log 2 - 2\pi i k} \neq 0. \quad (7.2)$$

The circle $|z| = R$ is a natural boundary of A .

Proof. Equation (6.4) shows that the radius of convergence is exactly R . Ignore the single correction at $n = 1$ for the moment. At $z = R\rho\omega_k$, formula (6.2) makes the summand $h(u_n)\omega_k^n \rho^n$. Since

$$\omega_k^n = e^{2\pi i k \alpha} e^{-2\pi i k u_n},$$

Lemma 6.1, applied to the Riemann integrable function $h(u)e^{-2\pi iku}$, gives the Cesàro mean

$$\begin{aligned} \lim_{M \rightarrow \infty} \frac{1}{M} \sum_{n=0}^{M-1} h(u_n) \omega_k^n &= e^{2\pi i k \alpha} \int_0^1 2^{u-\alpha} e^{-2\pi i k u} du \\ &= \frac{2^{-\alpha} e^{2\pi i k \alpha}}{\log 2 - 2\pi i k}. \end{aligned}$$

In the integral, $e^{\log 2 - 2\pi i k} - 1 = 1$. The resulting value is nonzero for every integer k . Lemma 7.1 proves (7.2). The omitted correction contributes $(1 - \rho)R\rho\omega_k$, which tends to zero.

If A admitted a holomorphic extension across the point $R\omega_k$, it would remain bounded along the inward radial segment near that point. Equation (7.2) contradicts this. Thus every $R\omega_k$ is a singular point. Irrationality of α makes the set of ω_k dense on the unit circle. An extension across any other point of $|z| = R$ would be defined in an open neighborhood containing some $R\omega_k$, again a contradiction. Hence no point on the circle admits analytic continuation through the circle. $\square$

The limit (7.2) describes a radial blowup. It does *not* assert that these are isolated poles or provide residues of a meromorphic continuation: the singularities are dense, and no such neighborhood of continuation is available.

7.3 The full generating function and recurrence consequences

Corollary 7.3. *The circle $|t| = \sqrt{R}$ is a natural boundary of F .*

Proof. Equation (5.4) makes F holomorphic for $|t| < \sqrt{R}$, with a removable singularity at zero. Suppose it extended through some point t_0 on the boundary. Since $t_0 \neq 0$, the map $t \mapsto t^2$ has a local analytic inverse near t_0 . Also $2t_0 + 1 \neq 0$: the only zero is $-1/2$, whereas $\sqrt{R} > 1/\sqrt{2} > 1/2$. Solving (5.4) for $A(t^2)$ would therefore provide an analytic extension of A through t_0^2 , contrary to Theorem 7.2. $\square$

A power series $H(z)$ is D-finite if it satisfies a nonzero homogeneous linear differential equation

$$p_d(z)H^{(d)}(z) + \cdots + p_0(z)H(z) = 0, \quad p_j(z) \in \mathbb{C}[z], \quad p_d \neq 0. \quad (7.3)$$

This is the standard terminology developed in [3]. Away from the finitely many zeros of p_d , a linear differential equation has analytic coefficients after division by p_d ; its local solutions continue along paths avoiding those zeros. An analytic germ satisfying (7.3) cannot therefore have a natural boundary on a finite circle.

Likewise, an algebraic germ has only finitely many possible finite singularities. For an irreducible polynomial equation $p(z, H) = 0$ of positive degree in H , exclude the zeros of its leading coefficient in H and of its discriminant in H ; the implicit-function theorem gives local analytic branches at the remaining points. This also contradicts a natural boundary.

Finally, a sequence (u_n) is P-recursive if it satisfies an eventual nonzero linear recurrence with polynomial coefficients in n . Such a recurrence implies a D-finite generating function. To recall the direct mechanism, let $\theta = z d/dz$. Then $\theta(z^n) = nz^n$, so sums of $p_j(n)u_{n+j}z^n$ can be written as differential operators in θ applied to $H(z) = \sum u_n z^n$, up to finitely many initial terms. After clearing powers of z , the recurrence becomes a nonzero linear differential equation with a polynomial right-hand side. Differentiating sufficiently many times makes that right-hand side zero. These observations and the two natural-boundary results prove all remaining conclusions of Theorem 1.1.

8 A complete classification for two bases

The counterexample is part of a family in which the rationality question can be answered exactly. Fix integers $2 \leq a < q$ and define the digit polynomial

$$D_a(y) = 1 + y + \cdots + y^{a-1}. \quad (8.1)$$

Use the interleaved exponents

$$G_{2j+1} = a^j, \quad G_{2j+2} = q^j, \quad (8.2)$$

and the products $\prod_{i=1}^N D_a(x^{G_i})$. This is precisely (1.2) with fixed $r = a - 1$. The exponent generating function is $t/(1 - at^2) + t^2/(1 - qt^2)$, and $G_N \rightarrow \infty$. Let $v_{a,q}(N)$ count the coefficients equal to one and put

$$F_{a,q}(t) = \sum_{N \geq 0} v_{a,q}(N) t^N.$$

Two integers $a, q > 1$ are *multiplicatively dependent* when $q^u = a^v$ for some positive integers u, v ; equivalently, $\log_q a$ is rational.

Theorem 8.1 (Two-base rationality classification). *For every pair of integers $2 \leq a < q$,*

$$F_{a,q}(t) \text{ is rational} \iff a \text{ and } q \text{ are multiplicatively dependent.} \quad (8.3)$$

When $q^u = a^v$ with $u, v \geq 1$, necessarily $v > u$, and

$$v_{a,q}(N + 2v) = a^{v-u} v_{a,q}(N) \quad (8.4)$$

for every sufficiently large N . In particular the reduced denominator of $F_{a,q}(t)$ divides $1 - a^{v-u} t^{2v}$.

8.1 Exact counts for a general rectangle

Write

$$Q_{m,n}^{a,q}(x) = \prod_{j=0}^{m-1} D_a(x^{a^j}) \prod_{j=0}^{n-1} D_a(x^{q^j}), \quad (8.5)$$

and let $U_{a,q}(m, n)$ be its number of coefficients equal to one. The first product is $1 + x + \cdots + x^{a^m-1}$. The second is supported, with coefficient one, on the a^n integers whose n base- q digits belong to $\{0, \dots, a-1\}$.

The sorted support is traversed by a base- a counter interpreted in base q . A carry through t trailing maximal digits produces the gap

$$g_t = q^t - (a-1) \sum_{j=0}^{t-1} q^j = \frac{(q-a)q^t + a-1}{q-1}. \quad (8.6)$$

It occurs

$$(a-1)a^{n-1-t} \quad (8.7)$$

times: there are $a-1$ choices for the incremented nonmaximal digit and a^{n-1-t} choices for the higher digits. Because $q > a$, the gap sizes are strictly increasing, beginning with $g_0 = 1$. A nonunit gap is preceded and followed by unit gaps, and the first and last gaps are unit, just as in Lemma 3.1.

Define the exact threshold

$$T_{a,q}(m) = \min\{t \geq 0 : (q-a)q^t + a - 1 \geq (q-1)a^m\} \quad (8.8)$$

$$= \left\lceil \log_q \left(\frac{(q-1)a^m - (a-1)}{q-a} \right) \right\rceil. \quad (8.9)$$

The ceiling expression is an identity over the real numbers; the integer inequality in (8.8) is the appropriate computational definition.

Proposition 8.2 (General rectangular formula). *For $m, n \geq 1$,*

$$U_{a,q}(m, n) = 2a^{\max(0, n - T_{a,q}(m))}. \quad (8.10)$$

The empty-product cases are $U_{a,q}(m, 0) = a^m$ and $U_{a,q}(0, n) = a^n$.

Proof. Use Corollary 2.2 with interval length a^m . If $T = T_{a,q}(m) < n$, the number of gaps of size at least a^m is

$$H = \sum_{t=T}^{n-1} (a-1)a^{n-1-t} = a^{n-T} - 1.$$

Otherwise $H = 0$. In either case $2 + 2H$ is (8.10). The boundary cases follow from unique base expansions. $\square$

For clarity, the factor 2 in (8.10) is not replaced by a : it counts the two uniquely covered endpoints associated with each large gap, together with the two outer boundary contributions.

8.2 Multiplicative independence forces nonrationality

The gaps satisfy

$$g_{t+1} = qg_t - (a-1). \quad (8.11)$$

For $m \geq 1$, if $g_t \geq a^m$, then

$$g_{t+1} \geq qa^m - (a-1) \geq a^{m+1},$$

because $(q-a)a^m \geq a-1$. Hence

$$T_{a,q}(m+1) - T_{a,q}(m) \in \{0, 1\} \quad (m \geq 1). \quad (8.12)$$

The restriction $m \geq 1$ is necessary for this assertion: for example, $T_{3,4}(0) = 0$ and $T_{3,4}(1) = 2$. From (8.9),

$$\frac{T_{a,q}(m)}{m} \longrightarrow \log_q a < 1. \quad (8.13)$$

For $n \geq 1$, the even subsequence is

$$v_{a,q}(2n) = 2a^{e_n}, \quad e_n = \max(0, n - T_{a,q}(n)).$$

By (8.12), the increments of e_n belong to $\{0, 1\}$; by (8.13), $e_n/n \rightarrow 1 - \log_q a$. If a, q are multiplicatively independent, this limit is irrational. Apply Lemma 4.1 with $C = 2$ and $b = a$ to conclude that the even-section generating function is not rational. Then $F_{a,q}$ is not rational either.

8.3 Multiplicative dependence gives an eventual recurrence

Suppose $q^u = a^v$ for positive integers u, v . Since $a < q$, necessarily $u < v$. Put

$$M = a^v = q^u, \quad K = \frac{q-1}{q-a}, \quad D = \frac{a-1}{q-a}.$$

Writing $m = v\ell + r$ with $0 \leq r < v$, the inequality defining $T_{a,q}(m)$ is

$$q^t \geq Ka^{v\ell+r} - D. \quad (8.14)$$

Set $t = u\ell + s$ and divide by M^ℓ to obtain

$$q^s \geq Ka^r - DM^{-\ell}. \quad (8.15)$$

For each residue r , let

$$s_r = \left\lceil \log_q(Ka^r) \right\rceil. \quad (8.16)$$

We have $q^{s_r-1} < Ka^r \leq q^{s_r}$. The positive distance between Ka^r and q^{s_r-1} is fixed, whereas $DM^{-\ell} \rightarrow 0$. Thus, for all sufficiently large ℓ , the least integer s satisfying (8.15) is exactly s_r . This remains true when Ka^r itself is an exact power of q ; the strict lower inequality is still available. Therefore

$$T_{a,q}(v\ell + r) = u\ell + s_r \quad \text{for all sufficiently large } \ell, \quad (8.17)$$

and, uniformly over the finitely many residues,

$$T_{a,q}(m + v) = T_{a,q}(m) + u \quad \text{for all sufficiently large } m. \quad (8.18)$$

Because $\log_q a < 1$, the maxima in (8.10) no longer truncate for either $m = n$ or $m = n + 1$ once n is large enough. Increasing the full product length by $2v$ increases both rectangular indices by v ; equation (8.18) then increases the exponent of a in (8.10) by $v - u$. This proves (8.4) on both parities. Multiplying the generating function by $1 - a^{v-u}t^{2v}$ leaves only finitely many nonzero coefficients, so it is a polynomial. This proves the rational direction and completes Theorem 8.1.

8.4 Rational controls and explicit examples

The classification is not simply a separation between equal and unequal bases. Different bases can be dependent: for example, $4^3 = 8^2$. Table 2 records several eventual recurrences used as computational controls.

For a particularly simple closed form, take $a = 2$ and $q = 2^s$ with $s \geq 2$. For $n \geq 1$, writing $n = s\ell + r$, $0 \leq r < s$, one has $g_\ell < 2^n \leq g_{\ell+1}$. Indeed, $g_\ell < q^\ell \leq 2^n$ when $\ell \geq 1$, while $g_0 = 1 < 2^n$ when $\ell = 0$; and

$$g_{\ell+1} = \frac{(q-2)q^{\ell+1} + 1}{q-1} \geq \frac{q^{\ell+1}}{2} \geq 2^n.$$

It follows that $T_{2,2^s}(n) = \lfloor n/s \rfloor + 1$ and

$$v_{2,2^s}(2n) = 2^{n - \lfloor n/s \rfloor} \quad (n \geq 0). \quad (8.19)$$

Summing by residue classes gives

$$\sum_{n \geq 0} v_{2,2^s}(2n) z^n = \frac{\sum_{j=0}^{s-1} 2^j z^j}{1 - 2^{s-1} z^s}. \quad (8.20)$$

In contrast, every pair $(2, q)$ with $q > 2$ not a power of 2 gives a counterexample with the original binary factors $1 + x^{G_i}$.

Table 2: Dependent-base controls. Each displayed recurrence is eventually valid; the verification suite checks every $40 \leq N \leq 200$.

a	q	Relation	Eventual recurrence
2	4	$4 = 2^2$	$v_{2,4}(N + 4) = 2v_{2,4}(N)$
2	8	$8 = 2^3$	$v_{2,8}(N + 6) = 4v_{2,8}(N)$
3	9	$9 = 3^2$	$v_{3,9}(N + 4) = 3v_{3,9}(N)$
4	8	$8^2 = 4^3$	$v_{4,8}(N + 6) = 4v_{4,8}(N)$
4	16	$16 = 4^2$	$v_{4,16}(N + 4) = 4v_{4,16}(N)$
8	16	$16^3 = 8^4$	$v_{8,16}(N + 8) = 8v_{8,16}(N)$
9	27	$27^2 = 9^3$	$v_{9,27}(N + 6) = 9v_{9,27}(N)$

9 Reproducible exact computations

The proof does not infer an infinite statement from a long prefix. The computations serve three narrower purposes: validating the indexing and exceptional cases, checking the finite combinatorial identities by independent representations, and making the derived integer sequences easy to reuse.

9.1 Three distinct computational representations

The module `code/mixed_base.py` implements three routes. Direct multiplication stores every coefficient and multiplies by the digit factors from their definition. An endpoint sweep independently enumerates the sparse digit support and adds $+1$ at each interval start and -1 at each interval end; it computes the histogram of *all* coefficient values. The closed formula computes only the threshold and the resulting power of a . The first method does not use the interval model, and the second does not use the carry-gap counting formula.

The checker compares entire histograms when feasible, not merely their coefficient-one totals. It also checks total degree and the mass identity

$$\sum_s [x^s] Q_{m,n}^{a,q}(x) = Q_{m,n}^{a,q}(1) = a^{m+n}.$$

The zero-coefficient histogram entry refers only to exponents between zero and the polynomial's degree; infinitely many trailing zero coefficients are never counted.

The dense grid uses $2 \leq a \leq 6$, $a + 1 \leq q \leq a + 5$, and $0 \leq m, n \leq 4$. The larger sweep grid uses $(a, q) = (2, 3)$, $1 \leq m \leq 24$, and $1 \leq n \leq 16$. The carry checks enumerate $1 \leq n \leq 6$ for the same 25 base pairs as the dense grid. The threshold tests cover $0 \leq m \leq 100\,000$ for $(2, 3)$ and $0 \leq m \leq 1000$ for 49 additional enumerated base-pair cases ($2 \leq a \leq 8$ and $a + 1 \leq q \leq a + 7$; this includes $(2, 3)$ again). The largest directly expanded polynomial has degree 269 815.

The supplied run completed successfully. The file `data/verification.json` records the counts, the actual elapsed time, and a SHA-256 digest of the dense-grid histograms. Timing is an observation about the supplied execution environment, not a performance guarantee.

9.2 Exact thresholds, not floating-point logarithms

The following compact function computes any full-prefix count in the family. It is an executable form of Proposition 8.2.

Table 3: Executed finite verification. All decisions use exact integers.

Check	Instances
Dense rectangle: direct histogram = sweep histogram; count = formula	625
Larger binary–ternary rectangle: sweep count and mass identities	384
Enumerated carry-gap multiplicities and adjacency	150
Actual interleaved prefixes $P_0, \dots, P_{24}$	25
Threshold compared with both adjacent integer powers	149 050
Exact floor-formula identities, $0 \leq n \leq 10\,000$	10 001
Input recurrence (1.5)	1 000
Even–odd identity (5.3)	1 000
Dependent-family recurrence, seven families and 161 indices each	1 127

```

def count_one_prefix(N: int, a: int = 2, q: int = 3) -> int:
    if N < 0 or not 2 <= a < q:
        raise ValueError("Need N >= 0 and integer bases 2 <= a < q")
    m, n = (N + 1) // 2, N // 2
    if n == 0:
        return a**m
    if m == 0:
        return a**n
    t, q_power = 0, 1
    target = (q - 1) * a**m
    while (q - a) * q_power + a - 1 < target:
        q_power *= q
        t += 1
    return 2 * a**max(0, n - t)

```

The full module also offers a streaming threshold computation that reuses a^m and $q^{T(m)}$. For fixed bases, the first M thresholds need $O(M)$ comparisons and multiplications by fixed integers, with $O(M)$ -bit working integers. In a bit-cost model with linear-cost fixed-integer multiplication, this gives an $O(M^2)$ upper bound on total bit operations and $O(M)$ working bits when results are streamed. Materializing all large output integers may itself require quadratic total space in bits.

For a single rectangle, direct dense expansion has a degree that grows exponentially in the sparse-digit length. The sweep instead uses $O(a^n)$ endpoint events and the implementation sorts them in $O(a^n \log(a^n))$ comparisons. The formula avoids both enumerations. These counts describe arithmetic or comparison operations; they do not treat arbitrarily large integer operations as constant-time in the bit-complexity discussion.

9.3 Package contents and reproduction

Run the verification suite from the project directory with

```
python code/verify.py
```

It requires only the Python standard library and is written for Python 3.9 or later. Do not run it with `-O`, which disables assertions. The supplied verification was executed using Python 3.13. The command

```
python code/mixed_base.py 40 --a 2 --q 3
```

prints the first 41 full-prefix counts. The CSV files contain indices 0 through 1000; a separate plain-text file uses the familiar two-column sequence format. No OEIS identifier is assigned or claimed.

The PDF is built with two passes of `pdflatex article.tex`; all needed figure PDFs are included. Recreating the figures with `python code/make_figures.py` additionally requires Matplotlib. The plots use floating-point arithmetic only for visualization; all verification decisions and all exported integer sequence values are exact. The README gives the complete directory map and reproduction commands.

10 What is settled, and what remains

The universal implication in (1.1)–(1.2) is false. The counterexample uses positive integer exponents tending to infinity, a fourth-order constant-coefficient recurrence for those exponents, fixed binary factors, and the simplest nonzero coefficient value $k = 1$. The output generating function not only fails to be rational: it has a natural boundary and is not D-finite. Within the specified two-base digit family, Theorem 8.1 gives a complete rationality classification.

Three boundaries of this result should be kept separate. First, the exponents in the counterexample are not eventually increasing. The same is true of the general interleaving when $a < q$: eventually $q^j > a^{j+1}$, so each sufficiently late even-to-odd step decreases. Reordering all exponents would preserve a finite product with a fixed multiset of factors, but it changes the prefix sequence and need not preserve the rationality of the exponent generating function. Thus this construction does not settle a separately imposed monotone version.

Second, the proofs count coefficients equal to one. They do not establish a corresponding classification for every fixed $k > 1$. For those values, multiple neighboring intervals and their overlaps matter; the isolated-gap identity alone no longer determines the answer.

Third, the classification concerns the family $G_{2j+1} = a^j$, $G_{2j+2} = q^j$ with digit set $0, \dots, a-1$. It is not a classification of all C-finite exponent sequences. The arithmetic mechanism suggests studying which recurrence families force periodic relative scales and which allow irrational threshold frequencies, but no claim about all such families is made here.

These are directions left untouched by this report, not assertions that each associated broader question has independently been verified to be open in the literature. The completed result is the explicit counterexample, its natural-boundary strengthening, and the exact two-base classification.

References

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